Diffusing Quantum Circuits for Computational Fluid Dynamics

Will Buziak — Department of Computer Science

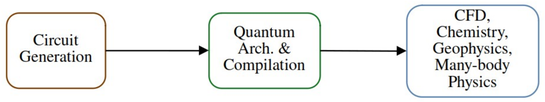
Iris Bahar — Department of Computer Science

Pooja Rao — NVIDIA

Abstract

Classical CFD modeling methods struggle to scale for real-world applications.

Quantum Architecture Search (QAS) and other machine-learning techniques can generate viable quantum circuits.



This work proposes a framework for quantum circuit generation in hopes to better utilize complex quantum modeling techniques on real quantum hardware.

Motivation

- Quantum-inspired CFD Algorithms have shown to be at least as efficient as classical Navier-Stokes simulations.
- Quantum Architecture methods such as gate cutting can further improve the fidelity of a quantum circuit and reduce depth.
- Quantum circuit generation mitigates the need for high-level algorithm expertise.
- An ML workflow with CUDA-Q at the center enables us to completely utilize the entire hardware stack for hybrid quantum-classical processing.

Challenges

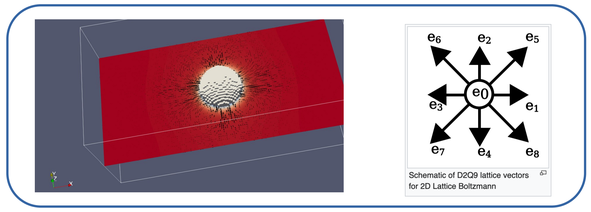
- Quantum Hardware is noisy & unreliable.
- Qubit technologies are non-homogeneous.
- Classical models experience exponential blowup when scaling to real-world systems.
- Algorithm design is non-trivial & often requires additional techniques to improve circuit fidelity on NISQ hardware.

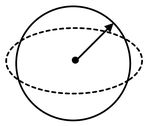
CFD

Classical modeling methods are **expensive** due to an inability to scale for:

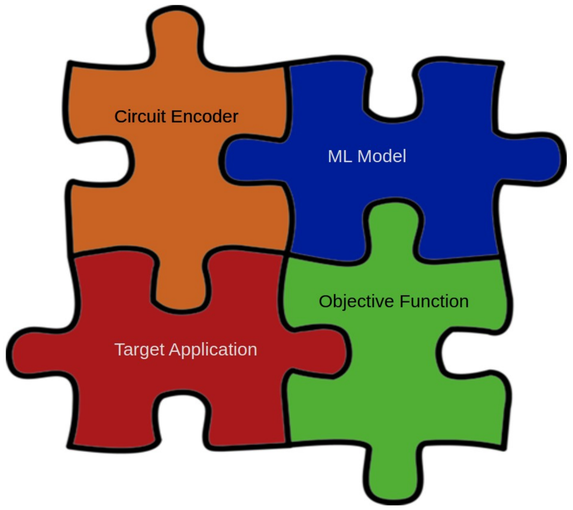
- Large / dense domain sizes
- Long-term model stability

We are interested in making quantum algorithms for modeling complex systems viable for **execution on real quantum hardware**.





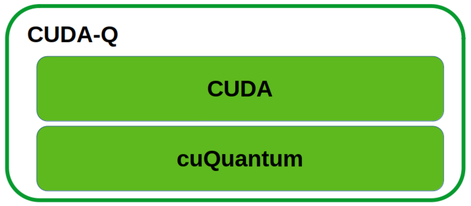
The target application dictates all other aspects of this framework.



CUDA-Q

NVIDIAs quantum SDK, CUDA-Q enables:

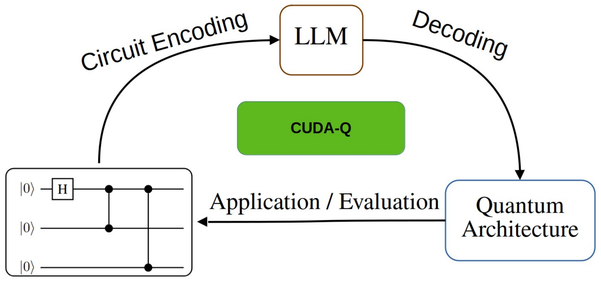
- Parallelization
- Hamiltonian Simulation
- Multi-GPU acceleration



CUDA-Q wraps NVIDIAs backend development kits, targeting high performance applications:

- cuQuantum: Library that accelerates quantum simulations on a circuit & device-level.
- CUDA: GPU development SDK

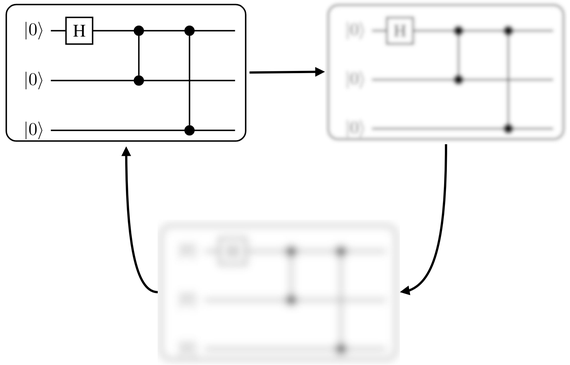
Ideal for hybrid quantum-classical processing and HPC applications.



Placing CUDA-Q at the center of this framework indicates the full utilization of hardware acceleration at every step.

Large-Language Models

Quantum circuit generation with diffusion models resembles image generation.



We imagine quantum circuits as a flexible representation - enabling us to consider them as images, or a language.

Insights

- The Evaluation metric can **prune** circuits with low fidelity, or inability to solve the target application.
- Each major component of this paradigm is intended to be **interchangeable**, opening the door for a larger swath of applications.
- Quantum computing provides speedups over classical solutions when algorithms can map to an algorithm that takes advantage of **entanglement similarities**.
- Quantum Architecture & Compilation frameworks can **target heterogeneous** distributed systems.

If we intend to make use of complex quantum algorithms, we need either:

- Scalable, reliable quantum hardware.

  or
- Sophisticated algorithm, architecture & compilation frameworks that enable execution on heterogeneous, distributed systems.